SE8205

MARIAN COLLEGE KUTTIKKANAM (AUTONOMOUS)

UG Programme (Honors) Degree March 2025 Regular Examinations

Second Semester

242CCRUBC102 Data Structures

Time: Maximum Marks: **60**

Answer one question from each bunch

BUNCH I

I.A While working on a student data management application, you are given an array A containing 8 elements indexed from 0, represented as A = [0, 1, 10, -1, -5, 100, 50, 20]. During a cleanup operation, you are instructed to remove the element at position 3. Demonstrate how this deletion can be performed, including how the elements are adjusted after the removal. Also, provide the algorithm used for deleting an element from a specific position in an array, explaining the steps involved in maintaining the structure of the array after deletion. (CO1:15 Marks)

OR

I.B While developing a system to manage a database of student records, you need to decide how to organize and store large volumes of data efficiently. In this context, explain the concept of a data structure and its importance in achieving efficient data organization and retrieval. Discuss the different classifications of data structures, including primitive, linear, and non-linear structures, and provide suitable examples for each. Further, illustrate how each type of data structure could be effectively applied to store and manage student records within the database.

(CO1:15 Marks)

BUNCH II

II.A In a scientific calculator application, a mathematical expression written in **infix form** needs to be converted into **postfix notation** for easier evaluation by the system. Consider the infix expression: **A + (B * C - (D / E - F) * G) * H.** Demonstrate the step-by-step conversion of this expression to postfix form using a **stack-based approach**, and display the **stack status after each step** in a tabular format. Also, provide the **algorithm** used for converting an infix expression to postfix notation.

(CO2:15 Marks)

OR

II.B.1 Riya is building a basic calculator application that uses a **stack** data structure with a maximum size of 6 to temporarily store numbers during operations. She performs the following set of stack operations in sequence: (i) Push(10), (ii) Push(0), (iii) Push(50), (iv) Pop(), (v) Pop(), and (vi) Push(100). Demonstrate how Riya executes each operation and show the status of the stack after every step. (CO2:6 Marks)

II.B.2 An embedded systems engineer is developing a custom expression evaluator for a microcontroller, where arithmetic expressions in postfix form must be evaluated efficiently using a stack. While testing the evaluator with the expression 8 2 3 ^ / 2 3 * + 5 1 * -, the engineer needs to determine the intermediate stack state for debugging purposes. Specifically, identify the top two elements of the stack immediately after the first multiplication (*) operation is executed. (CO2:6 Marks)

II.B.3 Comment on the statement "Stacks are known as LIFO Lists"

(CO2:3 Marks)

BUNCH III

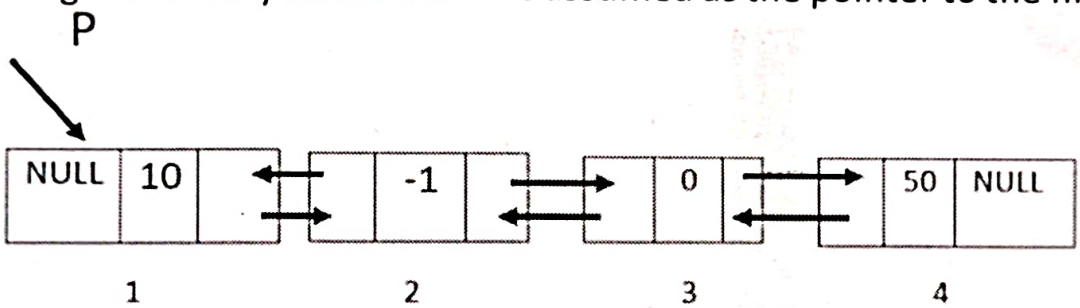
III.A.1 With an example, demonstrate the deletion of the first node of a given linked list . Also write the routine for deleting a node from a singly linked list. (CO3:10 Marks)

III.A.2 Consider a scenario where you are developing a program to manage a library system. As part of the system, you need to handle different types

of book genres such as fiction, non-fiction, science fiction, mystery, and so on. To enhance code readability and maintainability, you decide to use typedef to define custom data types for these genres. Describe how you would use the typedef statement in CPP to define custom data types for the book genres. Provide an example code demonstrating the usage of typedef for defining these custom data types. (CO3:5 Marks)

OR

III.B.1 With a routine, demonstrate the deletion of the node 't' at position 3 of the given doubly linked list. 'P' is assumed as the pointer to the first node.



(CO3:10 Marks)

III.B.2 Consider a scenario where you're tasked with designing a library management system for a university. In this system, you need to efficiently manage information about each book, including its title, author, publication year, and availability. One of the requirements is to implement a data structure to store and organize this information, considering the need for easy traversal in both forward and backward directions. Your task is to discuss the significance of self-referential structures in implementing doubly linked lists for this library management system. Explain how the use of self-referential structures facilitates the creation and manipulation of doubly linked lists to efficiently manage the collection of books. (CO3:5 Marks)

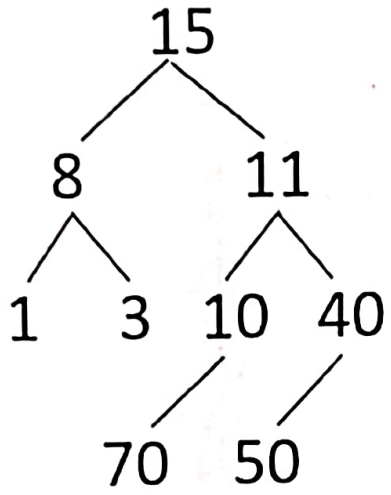
BUNCH IV

IV.A.1 Considering the binary tree given below, find out the order of the nodes visited in,

(i) inorder traversal

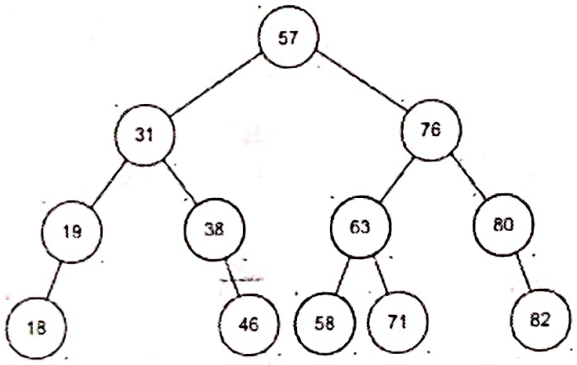
(ii) preorder traversal

(iii) postorder traversal



(CO4:6 Marks)

IV.A.2 Explain how many key comparisons are necessary to perform a search for the key 63 in the tree given below.



(CO4:5 Marks)

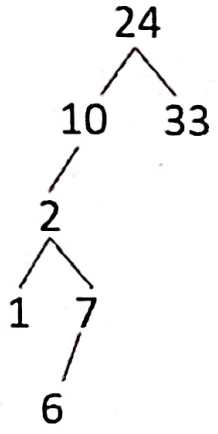
IV.A.3 Is it a valid structure definition that can be followed by a node in a binary tree. Justify your answer.

```
struct node
  {
   int data;
   struct node* left;
   struct node* right;
  };
```

(CO4:4 Marks)

OR

IV.B.1 Apply searching on the given binary search tree and decide whether the item 6 is present in the given tree. With suitable diagrams show the steps in performing the search.



(CO4:5 Marks)

IV.B.2 You are working as a software engineer at a tech company specializing in developing algorithms for data manipulation. Your team has been tasked with designing and implementing a binary search tree (BST) data structure. As part of the development process, you need to thoroughly understand how nodes are represented within the BST. Discuss and demonstrate the representation of a node in a binary search tree.

(CO4:5 Marks)

IV.B.3 Imagine you are tasked with explaining full and complete binary trees to a group of students. Your task is to create a diagram representing an example of such a tree and to develop a scenario-based question to assess their understanding. Draw a neat diagram representing an example of a full and complete binary tree. Ensure that the tree adheres to the properties of a full and complete binary tree. (CO4:5 Marks)

CO1	Illustrate the fundamental concepts of Data Structures, the Representation of single and two-dimensional arrays and the implementation of various operations on them
CO2	Analyse the representation of stacks and queues using arrays, operations on them and application of these data structures in problem-solving
CO3	Demonstrate the ability to implement and manipulate various types of linked lists (singly, doubly, and circular)